

MUHAMMED FARIS MUKTHAR M V

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AI/ML Engineer

 [MUHAMMEDFARISMUKTHARMV](#)

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Driven R&D Engineer with a Master's in Computer Science (Data Analytics), specializing in Machine Learning, AI, and IoT systems. Strong in Python, and MicroPython, with practical experience developing ML models for document automation, extraction, and computer vision tasks. Hands-on with building IoT prototypes using SBC's, MCU, and differnet sensors. Skilled in web development using Next.js, Node.js, HTML, CSS, and JavaScript to create responsive interfaces for real-time data and hardware integration. Known for solving real-world problems by combining hardware and software, developing efficient pipelines, and delivering innovative solutions with strong analytical and rapid prototyping skills.

SKILLS

Programming Languages	Python,SQL,HTML/CSS,JavaScript, Embedded C/C++
Communication	English, Malayalam(Mother Tongue), Hindi , Arabic
Developer Tools	Microsoft Excel,Microsoft Azure, PyCharm, Tableau, Power BI,
Web Development	Next.js, Node.js, REST APIs
Hardware Skills	Sensor integration, soldering, rapid prototyping

TECHNICAL EXPERIENCE

R&D Engineer – Innovations

Oct 2025 — Present

Digital University Kerala

Technopark Phase IV, Trivandrum

- Designing and developing end-to-end R&D prototypes integrating AI/ML, IoT hardware, and web technologies.
- ML model implementation on edge devices for detection, recognition, and smart automation.
- Real-time IoT data visualization and management through web dashboards.
- Working with single board computers and microcontrollers like Raspberry Pi, ESP32, and Arduino for rapid prototyping.
- Implementing different APIs for seamless communication between hardware, backend services, and user interfaces.
- Mentoring and guiding students to participate in hackathons, innovation challenges, and applied research projects.
- Supporting ML-driven ideation, project formulation, and technical execution of student-led innovation initiatives.
- Assisting in the planning, coordination, and conduct of academic and innovation programs, workshops, and technical events.
- Collaborating with faculty and institutional teams to promote an innovation-driven and research-oriented culture.

R&D Intern

Apr 2025 — Oct 2025

Cisco thinQbator-DUK

Technopark Phase IV,Trivandrum

- Developed and integrated IoT systems using Raspberry Pi, Arduino Uno/Nano, ESP32, and various sensors.
- Built a responsive interface using Next.js to stream and visualize sensor readings in real-time.
- Integrated ML models with Raspberry Pi for detections and identifications.
- Gained hands-on experience in soldering and physical assembly of embedded systems.
- Worked on rapid prototyping techniques for smart hardware integration.

Consultant – Machine Learning Engineering

Sep 2024 — Dec 2024

Cleareye.ai

Technopark Phase I,Trivandrum

- Designed and fine-tuned ML models for document classification, extraction, and prediction in the Trade Finance domain.
- Utilized RoBERTa with NLP techniques for noun detection and leveraged LayoutLM with OpenCV to enhance document analysis accuracy and processing efficiency..
- Implemented Siamese networks for signature verification, ensuring robust document validation.
- Created synthetic datasets to overcome data scarcity and improve model training quality.
- Applied object detection techniques like YOLO to validate document integrity and optimize processing workflows.

Machine Learning Intern

May 2024 – Aug 2024

Cleareye.ai

Technopark Phase I,Trivandrum

- Categorized documents using content and metadata classification techniques.
- Improved data extraction efficiency by fine-tuning and training LayoutLM for complex document layouts, tailored to meet specific requirements.
- Generated synthetic datasets to expand training samples and improve model robustness.

Machine Learning Research Intern

Feb 2024 – May 2024

Digital University Kerala

Technopark Phase IV,Trivandrum

- Document sanctity checks were performed using structural OCR and LayoutLM models.
- Validated document authenticity by enforcing YOLO-based detection for seals and signatures.
- Reduced manual verification efforts by automating structural and layout-based inspection

PROJECTS

Document Extraction Chatbot

Large Language Models

- Developed a government order chatbot using TensorFlow and NLP for fast, accurate query resolution.
- Using LangChain and LLMs, build context-aware responses, showcasing proficiency in deep learning for text processing and document understanding.

Plant Disease Detection

Computer Vision

- Leveraged the YOLOv8 models to classify plant health using leaf images and detect possible diseases.
- Enabled proactive measures by providing insights on potential diseases.

Model Prediction

Deep Learning

- Developed predictive models using CNN, RNN, and LSTM for the IMDB dataset and spam message datasets.
- Improved accuracy by optimizing neural network architectures and backpropagation.

Customer Centric Retail Analysis Trends and Time Series Forecast

Machine Learning

- Analyzed purchasing trends and built predictive purchase model.
- Delivered insights that are actionable to contribute to improved customer satisfaction in driving business growth..

EDUCATION

Master of Computer Science in Data Analytics, Digital University Kerala,Trivandrum	Aug 2022 – Jul 2024
Bachelor of Science in Mathematics, Farook College, Calicut University	Jun 2019 – Mar 2022
Science (PCMB), Farook HSS, Farook College	Jul 2017 – Mar 2019

ACTIVITIES

Company Qaurtermaster Sergeant - NATIONAL CADET CORPS - India	2021 — 2022
'C' Certificate holder - NATIONAL CADET CORPS - India.	2022